

Homework 2

Math 2031 — Professor Hamilton

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Section 2.1

Question 21

Statement: Let A and B be sets contained in universal set U . Which one(s) of the following statements are equivalent to the statement $A \subseteq B$? Give reasons for your answers.

- (A) $\forall x \in U, x \in A \wedge x \in B$. Not equivalent, this is the definition of intersection, $\cap$.
- (B) $\forall x \in U, x \notin B \rightarrow x \notin A$. Equivalent due to contrapositive, since A is a subset of B , if an element does not exist within B then we must conclude that it does not also exist within A .
- (C) $\exists x \in U \mid x \in A \wedge x \in B$. Not equivalent, as for $\subseteq$ this must apply $\forall x$.
- (D) $\forall x \in U, x \in A \rightarrow x \in B$. True by definition, since A is a subset of B , if $x \in A$ it must follow $x \in B$ by definition.
- (E) $\forall x \in U, x \notin A \rightarrow x \notin B$. False, let $A = \{1, 2, 3\}$ and $B = \mathbb{Z}$. $4 \in B \wedge 4 \notin A$ even though $A \subseteq B$.

Section 2.2

Question 1

Given: Let $A = \{1, 2, 3, 4, 5\}$, $B = \{0, 1, 4, 8\}$ and $C = \{2, 5, 7, 9, 11, 13, 17\}$. Compute each of the following:

- (A) $A \cup B = \{0, 1, 2, 3, 4, 5, 8\}$
- (B) $A \cap B = \{1, 4\}$
- (C) $A \cap C = \{2, 5\}$
- (D) $A - B = \{2, 3, 5\}$
- (E) $A - (B \cup C) = \{3\}$
- (F) $(A - B) \cup (A - C) = \{1, 2, 3, 4, 5\}$
- (G) $A \cap (B \cup C) = \{1, 2, 4, 5\}$
- (H) $(A \cap B) \cup (A \cap C) = \{1, 2, 4, 5\}$
- (I) $A \cup (B \cap C) = \{1, 2, 3, 4, 5\}$

(J) $(A \cup B) \cap (A \cup C) = \{1, 2, 3, 4, 5\}$

Question 2

Given: Let A, B and C be the sets given in the previous exercise and let the universal set $U = \{0, 1, 2, 3, 4, \dots, 20\}$. Compute the following:

Note: Here ‘...’ means sequential integers from a to b .

- (A) $\overline{A} = \{0, 6, 7, 8, \dots, 20\}$
- (B) $\overline{B} = \{0, 2, 3, 5, 6, 7, 9, \dots, 20\}$
- (C) $\overline{A \cap B} = \{0, 2, 3, 5, \dots, 20\}$
- (D) $\overline{A \cup B} = \{0, 2, 3, 5, 6, 7, \dots, 20\}$
- (E) $\overline{A \cup B} = \{6, 7, 9, \dots, 20\}$
- (F) $\overline{A \cap B} = \{0, 6, 7, 10, \dots, 20\}$

Question 14B

Statement: Let A and B be sets. Prove the following: $A \cup (A \cap B) = A$

Proof:

Let A and B be sets and assume $x \in A \cup (A \cap B)$. We must show that:

I.) $A \cup (A \cap B) \subseteq A$ and II.) $(A \subseteq A \cup (A \cap B))$

Part I:

Let $x \in A \cup (A \cap B)$. We must show that $x \in A$ for the definition of a subset. Since $x \in A \cup (A \cap B)$ by assumption, $x \in A \vee (x \in A \wedge x \in B)$, which necessitates that $x \in A$ in both cases, thus we have shown $\forall x \in A \cup (A \cap B), x \in A$.

Part II:

Let $x \in A$. We must also show that $x \in A \cup (A \cap B)$ for the definition of a subset. By assumption $x \in A$ and as per Part I, $x \in A \cup (A \cap B)$ necessitates that $x \in A \therefore \forall x \in A, x \in A \cup (A \cap B)$.

Thus $A \cup (A \cap B) = A$. ■

Question 17

Statement: Let A, B , and C be sets.

- (A) **Conjecture:** $A \subseteq B \cup C \rightarrow A \subseteq B \vee A \subseteq C$

Proof:

Let A, B, C be sets such that $A \subseteq B \cup C$.

We want to show that $A \subseteq B \vee A \subseteq C$, i.e., $\forall x \in A, x \in B \vee x \in C$, by the definition of subset and union.

By assumption since $A \subseteq B \cup C$, it follows that $\forall x \in A, x \in B \vee x \in C$, as per the definition of union.

Case 1: $x \in B$

If we assume $x \in B$ by the definition of union, and since $x \in A$ by our initial assumption, it must follow that $A \subseteq B$ which would satisfy one of the conditions of our conclusion, which is what we wanted to demonstrate.

Case 2: $x \in C$

Similar behavior to the case above.

Therefore, we have shown that $A \subseteq B \cup C \rightarrow A \subseteq B \vee A \subseteq C$. ■

(B) $A \not\subseteq B \wedge A \not\subseteq C \rightarrow A \not\subseteq B \cap C$

(C) **Conjecture:** $A \subseteq B \vee A \subseteq C \rightarrow A \subseteq B \cup C$

Proof:

Let A, B, C be sets such that $A \subseteq B \vee A \subseteq C$.

We want to show that $A \subseteq B \cup C$, i.e., $\forall x \in A, x \in (B \cup C)$, by the definition of subset and union.

By assumption since $A \subseteq B \vee A \subseteq C$, it follows that $\forall x \in A, x \in B \vee x \in C$.

Case 1: $A \subseteq B$

If $A \subseteq B$, it would mean that $\forall x \in A, x \in B$.

We want to show by the definition of union, $x \in (B \cup C)$.

Since we assumed $A \subseteq B$, it must follow that $x \in B$, moreover it must also be in $B \cup C$ as by definition of union, $x \in B \vee x \in C$.

Case 2: $A \subseteq C$

Same steps as above, now instead for C . Therefore, if $A \subseteq B \vee A \subseteq C$ it must follow that $A \subseteq B \cup C$. ■

Question 22B

Conjecture: $(A \cup B) - (A \cap B) = (A - B) \cup (B - A)$

Proof:

Part 1: Prove $(A \cup B) - (A \cap B) \subseteq (A - B) \cup (B - A)$

Let $x \in (A \cup B) - (A \cap B)$.

We want to show $x \in (A - B) \cup (B - A)$.

By assumption: $x \in (A \vee B) \wedge x \notin (A \wedge B)$ by definition of union and complement.

Using De Morgan: $x \in (A \vee B) \wedge (x \notin A \vee x \notin B)$. This creates 2 cases for us:

Case 1: $x \in A \wedge x \notin B$.

If $x \in A \wedge x \notin B$, it would also be in $(A - B) \cup (B - A)$, as by the definition of complement and union: $x \in A \wedge x \notin B = (A - B)$ and thus $x \in (A - B) \cup (B - A)$.

Case 2: $x \in B \wedge x \notin A$.

Same as case above however now we are working with $x \in B \wedge x \notin A = x \in (B - A)$.

Part 2: Prove $(A - B) \cup (B - A) \subseteq (A \cup B) - (A \cap B)$

Let $x \in (A - B) \cup (B - A)$.

We want to show $x \in (A \cup B) - (A \cap B)$.

Which again gives us 2 cases:

Case 1: $x \in (A - B)$.

Assume $x \in (A - B)$.

Since $x \in A$ it follows then that $x \in (A \cup B)$ by the definition of union.

Case 2: $x \in (B - A)$.

Since $x \in B$ it also follows that $x \in (A \cup B)$ by definition of union.

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Thus we have shown that $(A \cup B) - (A \cap B) = (A - B) \cup (B - A)$. ■

Question 23

Conjecture: $A \subseteq B \Leftrightarrow A \cap B = A$

Proof:

Part 1: $A \subseteq B \rightarrow A \cap B = A$

Assume $A \subseteq B$.

We want to show that $A \cap B = A$, i.e, $A \cap B \subseteq A \wedge A \subseteq A \cap B$.

1A) Proving $A \cap B \subseteq A$

Assume $x \in A \cap B$, meaning $x \in A \wedge x \in B$ by definition of intersection.

We want to show that $x \in A$ which is true by our assumption, thus $A \cap B \subseteq A$.

1B) Proving $A \subseteq A \cap B$

Let $x \in A$. We want to show that $x \in A \cap B$. By the definition of intersection, $x \in A \wedge x \in B$.

Moreover, we assumed that $A \subseteq B$, therefore also $x \in B$.

Thus $A \subseteq A \cap B$, which is what we wanted to show.

Part 2: $A \cap B = A \rightarrow A \subseteq B$

Assume $A \cap B = A$.

We want to show $A \subseteq B$.

By assumption, $A \cap B = A$ would mean that $A \cap B \subseteq A \wedge A \subseteq A \cap B$.

We want to show that $\forall x \in A, x \in B$.

By assumption, $A \subseteq A \cap B$, meaning $\forall x \in A, x \in A \wedge x \in B$, thus $x \in B$ would necessitate that $A \subseteq B$, which is what we wanted to show.

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Therefore: $A \subseteq B \Leftrightarrow A \cap B = A$. ■

Question 25A

Conjecture: $A - (B \cup C) = (A - B) \cap (A - C)$

Proof:

Note: For this proof, I am proving the sets are equal by their logical operators, moreover since the proof is bidirectional, it works both ways.

Assume $x \in A - (B \cup C)$.

We want to show $x \in (A - B) \cap (A - C)$.

By assumption: $x \in A \wedge (x \notin B \wedge x \notin C)$ using De Morgan's Law.

By distributive property, rewrite our assumption as $(x \in A \wedge x \notin B) \wedge (x \in A \wedge x \notin C)$.

And going back using the definition of complement and intersection: $(A - B) \cap (A - C)$.

Since every step in the proof was biconditional due to the logical equivalence of the sets, our argument works in both direction of proving the equality of the sets.

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Thus $A - (B \cup C) = (A - B) \cap (A - C)$. ■